Changing matter

Changes to matter can be classified as either **physical** or **chemical** changes. During a **physical change** some of the characteristics of the substance change, but the substance remains the same. No new substance is formed.

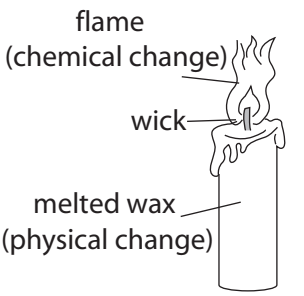
When we boil water, it changes into steam. Steam is another form of water. When water is heated, this does not create a new substance – it simply changes the water's state from a liquid to a gas. Examples of physical changes to matter include:



- changes of state (solid, liquid, gas)
- changes in shape or size
- dissolving.

Physical changes are usually easy to change back, or **reverse**. For example, if you dissolve salt in water, you can separate the salt back out by boiling the water.

When a **chemical change** occurs, the original substance is changed into something new. A chemical change is caused by a chemical reaction. A chemical change is **irreversible** – you cannot get the original substance back. For example, when a candle is lit, some of the wax melts, which is a physical change, and some of it burns and changes to ash and smoke, which is a chemical change.



To identify that a chemical change has taken place, we can look for the following clues:

- The substance fizzes or bubbles (gas is produced).
- Smoke is produced.
- Heat is produced.
- Light energy is produced (the substance glows, or a flame is produced).
- You cannot get the original substance back.
- There is a permanent colour change.
- There is a smell.
- A solid forms when two liquids are mixed.

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Physical and chemical changes

Read the Resource sheet 'Changing matter', then answer the following questions.

☞ **1** Unjumble the sentence parts and then rewrite each sentence about physical and chemical changes in the correct order.

are physical changes. that do not produce Changes in matter new substances

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is an A change of state physical change. example of a

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reversible. changes are Most physical

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is an example Burning the wick of a candle chemical change. of a

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new substance change occurs a When a chemical is formed.

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2 Identify which changes in the table below are physical changes and which are chemical changes. For chemical changes, complete the clue that shows it is a chemical change. The first one has been done for you.

Change to matter	Physical or chemical change	Clue that shows it is a chemical change
fireworks going off	chemical change	heat and light produced
water freezing		
lighting a match		
cooking a cake		
baking soda in vinegar		
crushing a piece of chalk		
melting chocolate		

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